

HIGH ACRYLONITRILE NBR: ORGANIC-SOLVENT FREE-RADICAL AND RELATED COPOLYMERIZATION ROUTES

1. ABSTRACT

This report synthesizes patent evidence relevant to high acrylonitrile nitrile butadiene rubber (NBR), with emphasis on solution-based and composition-controlled routes that can support elevated acrylonitrile incorporation. The target material is defined as NBR with increased acrylonitrile or bound acrylonitrile content relative to standard grades, typically achieved by controlling the acrylonitrile/butadiene monomer feed, conversion, and regulator level during copolymerization. Across the patent set, the dominant disclosed chemistry is free-radical copolymerization of at least one conjugated diene with at least one α,β -unsaturated nitrile in organic solvent, often with optional termonomers and molecular-weight regulators. One patent focuses on solution polymerization in specific solvent mixtures, while another discloses solution polymerization in organic solvents using RAFT-type regulators and staged initiator addition. A third patent, although not an NBR disclosure, provides a useful comparator for controlled solution copolymerization in organic media using a neodymium Ziegler-Natta catalyst system and demonstrates how feed composition and catalyst design can be used to tune diene copolymer microstructure and incorporation behavior.

The landscape shows two major process trends for high-acrylonitrile NBR: first, organic-solution polymerization as an alternative to conventional emulsion polymerization, intended to avoid latex isolation before downstream hydrogenation and to improve process integration; second, deliberate control of monomer feed and reaction time to manage conversion, molecular weight, and composition drift. The EP and US NBR patents both emphasize organic solvents, free-radical initiation, and molecular-weight control, with one explicitly using dodecyl mercaptan-type regulators in the background and the other using a RAFT regulator (DoPAT) with Vazo 88 initiator. The examples show acrylonitrile charges around 37 phm in one case and controlled conversion endpoints of 39% and 48% in others, indicating that composition and conversion are central levers. The comparator patent demonstrates solution polymerization with 1-phenylbutadiene/butadiene using a neodymium catalyst system, highlighting the broader relevance of controlled monomer feed and catalyst selection for diene copolymer synthesis.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2478021B1
- Patent Title: Nitrile rubbers and production of same in organic solvents
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2015
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical solution polymerization of nitrile rubbers in organic solvents, using at least one conjugated diene, at least one alpha,beta-unsaturated nitrile, and optionally additional copolymerizable monomers. The examples show acrylonitrile and 1,3-butadiene polymerized in dimethylacetamide with Vazo 88 and DoPAT, with optional termonomers such as hydroxyethyl acrylate, styrene, and HDODA, followed by precipitation in ethanol. - Vazo 88 loading: 284.9 mg (1.17 mmol, 0.31 phm) — Example 1 - DoPAT loading: 204 mg (0.583 mmol, 0.22 phm) — Example 1 - Dimethylacetamide: 50 mL (51 phm) — Example 1 - Acrylonitrile: 43 mL (653.3 mmol, 37 phm) — Example 1 - 1,3-Butadiene: 90 mL (1077 mmol, 63 phm) — Example 1 - Butadiene pressure: 3 bar — Example 1 - Polymerization temperature: 100°C — Example 1 - Polymerization time: 9 hours — Example 1 - Final conversion: 39% — Example 6 - DoPAT loading: 2.22 phm — Example 7 - Polymerization time: 9 h — Example 7 - Final conversion: 48% — Example 7 - Styrene: 2.79 g (26.8 mmol, 50 phm) — Example 8 - Polymerization temperature: 100°C — Example 8 - Polymerization time: 13 hours — Example 8 - Final conversion: 9.2% — Example 8 - Vazo 88 post-dose: 128 mg (0.523 mmol, 0.15 phm) — Examples 18, 20 and 23 - Post-dose time: 2.5 hours and 6 hours — Examples 18, 20 and 23 - Butadiene pressure: 3 bar — Examples 17-23 - Polymerization temperature: 100°C — Examples 17-23 - Polymerization time: 9 hours — Examples 17-23

Relevant Experimental Evidence - Example 1: Demonstrated baseline solution polymerization of acrylonitrile/butadiene in DMAc at 100°C and 3 bar butadiene, with isolation by ethanol precipitation. - Example 6: Demonstrated terpolymer formation by adding hydroxyethyl acrylate, with the reaction stopped after 6 hours at 39% conversion. - Example 7: Demonstrated deliberate low-molecular-weight NBR preparation by increasing DoPAT to 2.22 phm and stopping at 48% conversion after 9 h. - Example 8: Demonstrated chain extension/blocking of an NBR RAFT macromer with styrene at 100°C. - Examples 17-23: Demonstrated solution polymerization with acrylonitrile, butadiene, optional HDODA/HEA, and staged initiator addition in selected runs.

Technical Relevance This patent is directly relevant because it discloses organic-solvent free-radical synthesis of NBR from acrylonitrile and butadiene, including composition control through feed amounts, regulator level, and conversion. It is one of the clearest sources for

high-acrylonitrile NBR preparation in solution rather than conventional emulsion polymerization.

Patent 2

- Patent Number: US20140235744A1
- Patent Title: Process for producing nitrile rubbers in organic solvents
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2014
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses free-radical polymerization of nitrile rubbers in organic solvents using at least two solvents, with the main solvent comprising 70 to 99.9 vol% of the solvent mixture. The process may be run without a molar-mass regulator or with a regulator compound, and the disclosure emphasizes improved time-conversion behavior for solution polymerization of acrylonitrile/butadiene systems. - Main solvent amount: 70 to 99.9 vol% of total solvents — Claim 1 - Reaction temperature: 343 K — cited prior-art comparison in description - Reaction time: 40 hours — cited prior-art comparison in description - Conversion: 40.5% — cited prior-art comparison in description - Reaction temperature: 353 K — cited prior-art comparison in description - Reaction time: 18 hours — cited prior-art comparison in description - Conversion: greater than 40% — cited prior-art comparison in description

Relevant Experimental Evidence - No numbered examples were detected by the parser; the disclosure is primarily claim- and description-based. - The patent emphasizes that solution polymerization in organic solvent can achieve improved time-conversion curves relative to earlier literature. - The claims define a two-solvent system with a dominant main solvent fraction, which is the key process distinction.

Technical Relevance This patent is relevant because it addresses the same target chemistry, nitrile rubber solution polymerization, and provides a solvent-mixture strategy that may improve processability for high-acrylonitrile grades. Although explicit high-ACN example data were not extracted, the process framework is directly applicable to acrylonitrile-rich NBR synthesis.

Patent 3

- Patent Number: EP2727948B1
- Patent Title: High cis diene/phenylbutadiene copolymers prepared using a ziegler/natta ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown

- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses solution polymerization of 1,3-butadiene with substituted butadienes using a neodymium-based Ziegler-Natta catalyst system formed from an organoaluminum compound, a neodymium carboxylate, and a dialkyl aluminum chloride. It is not an NBR patent, but it is relevant as a controlled solution copolymerization reference showing how monomer feed, catalyst composition, and solvent choice affect incorporation and microstructure. - Butadiene solution concentration: 15% in hexanes — Example 2 - 1-Phenylbutadiene loading: 48 mg (1.2 weight % relative to butadiene) — Example 2 - TIBA solution: 0.8 mL of 0.1 M in hexanes — Example 2 - Catalyst solution: 0.42 mL of 0.0191 M 1/25/2.5 Nd Versatate/TIBA/DEAC — Example 2 - Polymerization temperature: 65 °C — Example 2 - Polymerization time: 1 h — Example 2 - Termination solution: 0.5 mL of 10% BHT/isopropanol — Example 2 - Polymer yield: 90% — Example 2 - Mn: 162,000 — Example 2 - PDI: 2.5 — Example 2 - 1-Phenylbutadiene feed levels: 2.4 and 11 weight percent relative to butadiene — Example 2 - Reaction time: 2 h — Table 2 - Reaction time: 9 h — Table 2 - Mw: 515×10^3 — Table 2 - Mn: 200×10^3 — Table 2 - PDI: 2.6 — Table 2 - Mw: 400×10^3 — Table 2 - Mn: 162×10^3 — Table 2 - PDI: 2.5 — Table 2 - Mw: 398×10^3 — Table 2 - Mn: 163×10^3 — Table 2 - PDI: 2.4 — Table 2 - Mw: 327×10^3 — Table 2 - Mn: 114×10^3 — Table 2 - PDI: 2.9 — Table 2 - 1,4-cis microstructure: 98.0% — Table 3 - 1,4-trans microstructure: 1.5% — Table 3 - 1,2 microstructure: 0.5% — Table 3 - Catalyst loading: 0.2 mmphm NdVers/TIBA/DEAC 1:25:2.5 — Figure 2 and Figure 3 - Polymerization medium: 75 to 99.5 weight percent butadiene monomer — description - Phenylbutadiene monomer: 0.5 to 25 weight percent — description - Neodymium metal catalyst level: 0.05 to 1.0 millimole per 100 grams of monomer — description - Preferred neodymium metal catalyst level: 0.1 to 0.3 millimole per 100 grams of monomer — description - Catalyst preformation temperature: 0°C to 100°C — description - Preferred catalyst preformation temperature: 15°C to 30°C — description

Relevant Experimental Evidence - Example 1: Demonstrated synthesis of 1-phenylbutadiene precursor used for copolymerization. - Example 2: Demonstrated butadiene/1-phenylbutadiene copolymerization at 65°C for 1 h with a neodymium catalyst system, giving 90% yield and Mn 162,000. - Table 2: Demonstrated that increasing phenylbutadiene feed reduced reaction rate and altered molecular weight/yield. - Table 3: Demonstrated a highly cis-1,4 copolymer microstructure with 98.0% cis content. - Figures 2-3: Demonstrated catalyst-controlled copolymerization behavior across different substituted butadiene feeds.

Technical Relevance This patent is not a nitrile rubber disclosure, but it is technically relevant as a solution copolymerization benchmark for controlled diene incorporation and microstructure control. Its catalyst and feed-control concepts are useful comparators when evaluating process strategies for high-acrylonitrile NBR synthesis.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2478021B1 and US20140235744A1 both disclose free-radical nitrile rubber synthesis in organic solvent, but EP2478021B1 provides concrete acrylonitrile/butadiene example charges and RAFT-type control, whereas US20140235744A1 emphasizes a two-solvent system and improved time-conversion behavior.
- EP2478021B1 uses Vazo 88 and DoPAT with solution polymerization in DMAc at 100°C and 3 bar butadiene; US20140235744A1 does not extract example initiator data, but claims a main-solvent fraction of 70 to 99.9 vol% and discusses prior-art conversion limits around 40% over 40 h.
- EP2478021B1 explicitly reports acrylonitrile at 37 phm and butadiene at 63 phm in Example 1, supporting a high-ACN feed strategy; US20140235744A1 does not expose explicit monomer ratios in the extracted evidence.
- EP2478021B1 includes optional termonomers such as HEA and HDODA and staged initiator addition in selected examples, indicating broader composition control; US20140235744A1 focuses more on solvent architecture than on termonomer design.
- EP2727948B1 is mechanistically different, using a neodymium Ziegler-Natta catalyst system for diene copolymerization rather than nitrile-rubber free-radical chemistry, but it illustrates how solution polymerization, catalyst composition, and feed level can control incorporation and microstructure.
- Among the three, EP2478021B1 is the most directly applicable to high-acrylonitrile NBR because it explicitly polymerizes acrylonitrile with butadiene in solution and reports conversion, temperature, pressure, and regulator levels.

4. CONCLUSION

The patent evidence supports organic-solution free-radical copolymerization as the most relevant route to high-acrylonitrile NBR, with composition control achieved by acrylonitrile/butadiene feed ratio, regulator selection, and conversion management. EP2478021B1 is the strongest direct disclosure, showing acrylonitrile at 37 phm, butadiene at 63 phm, 100°C polymerization, 3 bar butadiene, and RAFT-type control in DMAc, while US20140235744A1 reinforces the value of solvent-mixture design for improved time-conversion performance. Based on the evidence, a practical high-ACN NBR synthesis window would center on solution polymerization in an organic solvent, controlled monomer feed with elevated acrylonitrile fraction, and careful molecular-weight regulation; staged initiator addition and optional termonomers can be used when broader property tuning is required.

5. REFERENCES

- EP2478021B1 | Nitrile rubbers and production of same in organic solvents | Not disclosed | EP | 2015 | <https://patents.google.com/patent/EP2478021B1/en>
- US20140235744A1 | Process for producing nitrile rubbers in organic solvents | Not disclosed | US | 2014 | <https://patents.google.com/patent/US20140235744A1/en>

- EP2727948B1 | High cis diene/phenylbutadiene copolymers prepared using a ziegler/natta ... | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2727948B1/en>